

NOS Practice Problem Statements

Spec-driven development practice — Beginner

How this works: pick a problem statement, write a short spec before writing any code, build a working prototype against that spec, then walk through a quick live demo.

Beginner

Single entity, CRUD plus one derived view (a total, a streak, a progress bar). Focus: clean spec, clean basic logic.

1. Nutrition Tracker

Required features:

- Log meals with calorie/macro values
- Show running daily calorie total
- View 7-day history

Edge case to handle:

What happens when a user logs a meal for a past date, not today?

2. Habit Tracker

Required features:

- Add habits
- Mark each habit done/skip per day
- Show current streak count per habit

Edge case to handle:

How does the streak behave after a skipped day — reset to 0 or pause?

3. Water Intake Reminder

Required features:

- Log intake amount
- Set a daily goal
- Show visual progress toward goal
- Reset progress at midnight

Edge case to handle:

What happens if the user logs intake after midnight but before checking the app — does it count for the new day or the old one?

4. Book Log

Required features:

- Add books with status (want/reading/done)
- Track pages read
- Show progress bar per book

Edge case to handle:

What happens if pages read exceeds total page count entered?

5. Study Timer

Required features:

- Start/stop timed sessions
- Log duration and subject per session
- Show weekly total hours per subject

Edge case to handle:

What happens if the app is closed mid-session — is the partial time logged or lost?

6. To-Do List with Priority

Required features:

- Add tasks
- Set priority (high/medium/low)
- Sort list by priority
- Mark complete

Edge case to handle:

How are completed tasks displayed — hidden, struck through, or moved to a separate list?